

Exact Quantum Values and Permutation-Blind Maximizers in Cyclic Bell Inequalities

Sharp operator bounds, equality structure, and limits of Bell-value randomness certification

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Abstract

Perito, D’Avino, Jung, Mironowicz, Acín, and Augusiak introduced two cyclic Bell families designed for efficient certification of maximal global randomness and conjectured the exact value of the reduced operator underlying the first family. We prove the conjectured value in every dimension and strengthen the upper bound to the commuting-operator model. The proof uses a polar-decomposition positive factor and a sharp scalar extremum on the unit circle. We then study the equality mechanism and identify a permutation freedom invisible to every first-harmonic correlator in the Bell score. For every $d \geq 4$, a final-two phase swap gives admissible order- d , rank-one projective measurements that attain the exact maxima of both augmented cyclic families while producing a nonuniform designated joint output distribution. Hence the maximum scalar Bell value alone does not certify $2 \log_2 d$ bits of global randomness across the whole maximizing face. This does not challenge randomness certification conditioned on the complete canonical behavior, nor does it show that the canonical strategy lacks maximal randomness. We give an exact $d = 4$ certificate over $\mathbb{Q}(\zeta_{16})$, all-dimensional analytic constructions, and scoped consequences for low-setting designs.

Status. This is an unrefereed, AI-assisted manuscript. Its proofs and verification artifacts are public for specialist audit; internal adversarial review and passing code are not peer review.

1 Introduction

Bell inequalities can certify randomness without a model of the devices, but the conditioning data matter. A complete behavior $p(a, b \mid x, y)$ contains far more information than one scalar Bell value. If a Bell maximum exposes a unique behavior, symmetries of the score may force uniform outputs and, with an operator-level argument, privacy against a purifying adversary. If the maximizing face contains several behaviors, the same symmetry argument need not apply to each of them [2, 3, 4].

Perito, D’Avino, Jung, Mironowicz, Acín, and Augusiak recently introduced two families built around cyclic Fourier correlators [1]. They defined the families, found explicit maximally-entangled strategies, proved the exact value of the second reduced family by a sum-of-squares identity, and conjectured the exact value of the first reduced family. Their symmetry analysis motivated maximal-global- randomness statements; for selected dimensions they also evaluated Eve’s guessing probability numerically after fixing the complete canonical behavior. These are logically different inputs to a randomness problem.

Topic	Originating work [1]	This paper
Bell families	Introduced both cyclic families and their normalization	Adopts the displayed Hermitian operator definitions, with a convention note
Attaining strategies	Supplied the canonical strategies and matching lower bounds	Rechecks admissibility and attainment exactly
First reduced value	Conjectured $2 \csc(\pi/(2d))$; numerical evidence through $d = 6$	Proves the all-dimensional upper bound and the $q = qa = qc$ equality
Second reduced value	Proved value d by an SOS identity	Uses that SOS, including its commuting-operator reading
Equality phases	Not a classification of the maximizing face	Gives a conditional phase-permutation mechanism and its exact product hypotheses
Alternative maximizers	Not identified	Constructs biased exact maximizers of both augmented families for every $d \geq 4$
Canonical full behavior	Randomness studied numerically in selected dimensions; second family self-tested at $d = 3$	Not contradicted
Scalar-value implication	Maximal-violation certification proposed, with a uniqueness-based symmetry rationale	Shows that the scalar maximum alone is insufficient for $d \geq 4$

Table 1: Attribution and logical scope. The present work does not claim the family definitions, canonical strategies, or second-family SOS.

The surrounding literature includes high-dimensional and modular Bell inequalities [6, 7, 8], self-testing from few measurements [9, 12], and analytic or algorithmic SOS constructions [11, 13]. Two contemporaneous works of the originating collaboration pursue all-dimensional global randomness and three-outcome noise robustness by different constructions [14, 15]. We use these works to set the attribution boundary; the family-specific polar bound and the phase-permuted maximizers below are not imported from them. To our knowledge, no prior work establishes either the first-family upper bound in this normalization or the nonuniform exact maximizers constructed here; the accompanying priority audit records the qualified search basis for that statement.

This paper unifies the exact-value and randomness questions through the equality phases of the first operator. The polar proof reduces its upper bound to a scalar roots-of-unity extremum. At equality the relative unitary may carry the full set of maximizing phases in different cyclic orders. Those orders leave the Bell-visible first harmonics unchanged, but they alter higher Fourier data and can change a complete output table. The same Bob observables also annihilate the published sum-of-squares factors for the second family.

Our main conclusions are as follows.

1. The finite-dimensional tensor-product, approximate, and commuting-operator values of the first reduced cyclic operator are all $2 \csc(\pi/(2d))$.
2. A polar-linear permutation theorem gives explicit sufficient conditions under which phase permutations remain admissible maximizers and preserve the entire first-harmonic correlator matrix.
3. For each $d \geq 4$, one such maximizer has uniform local marginals but a nonuniform designated joint table. With trivial Eve this already gives a guessing probability strictly larger than $1/d^2$.

4. The same construction maximizes the second augmented cyclic family and has the same target table.
5. The result separates scalar Bell-value certification from certification conditioned on a fixed full behavior. It is not a claim about the exact worst-case guessing probability, the complete maximizing face, or the randomness of the canonical realization.

The equality mechanism is proved in Sections 3–4; the two biased families are treated in Sections 5 and 6; and the randomness interpretation is isolated in Section 7. Section 8 records only the setting-complexity consequences that survive an independent audit.

2 Bell operators, behaviors, and randomness notions

Fix $d \geq 2$, write $\omega = e^{2\pi i/d}$, and identify outcomes with $\mathbb{Z}_d$. A projective d -outcome measurement $\{M_a\}_{a \in \mathbb{Z}_d}$ is encoded by the order- d unitary

$$A = \sum_{a=0}^{d-1} \omega^a M_a, \quad M_a = \frac{1}{d} \sum_{k=0}^{d-1} \omega^{-ak} A^k.$$

Zero spectral projections are allowed in the general definition. All explicit strategies below have the complete simple spectrum of d -th roots, so their outcomes are rank-one projectors.

For finite-dimensional tensor-product strategies, let q denote the supremum of a Bell functional, and let qa denote the value over the closure of those correlations. In the commuting-operator model qc , Alice’s and Bob’s PVMs act on one Hilbert space and commute across parties. We use

$$\mathcal{Q}_q \subseteq \mathcal{Q}_{qa} \subseteq \mathcal{Q}_{qc}.$$

The operator inequalities below hold in the commuting model; a matching finite-dimensional strategy then identifies all three values.

For a behavior $p(a, b \mid x, y)$, a scalar Bell maximum fixes only one supporting hyperplane of the quantum correlation set. We distinguish:

- the value of a Bell functional;
- its full maximizing face;
- a selected canonical strategy and its canonical behavior;
- the complete observed behavior, including higher Fourier correlators;
- uniform observed target probabilities; and
- private randomness against a purifying adversary.

For a purification $|\Psi\rangle_{ABE}$ and target settings (x_*, y_*) , let

$$\sigma_E^{ab} = \text{Tr}_{AB}[(M_a^{x_*} \otimes N_b^{y_*} \otimes \mathbf{1})|\Psi\rangle\langle\Psi|(M_a^{x_*} \otimes N_b^{y_*} \otimes \mathbf{1})].$$

Eve’s guessing probability is

$$G(AB \mid x_*, y_*, E) = \max_{\{Q_{ab}\}} \sum_{a,b} \text{Tr}(Q_{ab} \sigma_E^{ab}).$$

Maximal private global randomness means $\sigma_E^{ab} = \rho_E/d^2$ for every a, b , and implies $G = 1/d^2$. Uniform probabilities are necessary but not sufficient for this operator-valued condition. Conversely, one explicit nonuniform target table, even with one-dimensional Eve, is enough to rule out maximal private randomness for that realization.

Remark 2.1 (Normalization convention). The current source version of Ref. [1] has a localized normalization discrepancy: the displayed reduced value in its Eq. (17) and its $d = 3$ augmented value agree with the operator definition, while the formula printed beside its first augmented randomness conjecture contains an additional factor d in the denominator. We take the displayed Hermitian operator as authoritative. Thus the reduced and augmented values in our convention are respectively

$$M_d = 2 \csc\left(\frac{\pi}{2d}\right), \quad M_d + 1.$$

Appendix E records the corresponding adjoint conventions. Nothing in our argument uses the typographical discrepancy.

3 Exact value of the first cyclic operator

For commuting Alice and Bob unitaries define

$$\mathcal{I}_d = \sum_{y=0}^{d-1} \operatorname{Re}[(A_0 + \omega^y A_1) B_y]. \quad (1)$$

In the tensor model the product in (1) means a tensor product. The order- d relations are part of the Bell scenario, but will not be needed for the upper bound.

Theorem 3.1 (Sharp commuting-operator value). *Let $A_0, A_1, B_0, \dots, B_{d-1}$ be unitaries on an arbitrary complex Hilbert space and suppose $[A_x, B_y] = 0$ for every x, y . Then*

$$\mathcal{I}_d \leq M_d \mathbf{1}, \quad M_d = 2 \csc\left(\frac{\pi}{2d}\right). \quad (2)$$

After imposing the order- d measurement relations,

$$\boxed{\beta_q(\mathcal{I}_d) = \beta_{qa}(\mathcal{I}_d) = \beta_{qc}(\mathcal{I}_d) = M_d} \quad (d \geq 2).$$

The proof has two independent ingredients. The first is an operator identity that remains valid when a polar factor has a kernel.

Lemma 3.2 (Polar positive-factor identity). *Let $\mathcal{A}, \mathcal{B} \subseteq \mathcal{B}(\mathcal{H})$ be commuting unital $*$ -algebras, let $C \in \mathcal{A}$, let $C = V|C|$ be its canonical polar decomposition, and let $B \in \mathcal{B}$ be unitary. Put*

$$P_{C,B} = |C^\dagger|^{1/2} - V|C|^{1/2}B.$$

Then

$$\frac{|C| + |C^\dagger|}{2} - \operatorname{Re}(CB) = \frac{1}{2} P_{C,B}^\dagger P_{C,B}. \quad (3)$$

Proof. The initial projection $V^\dagger V$ is the support projection of $|C|$, while polar functional calculus gives

$$|C^\dagger|^{1/2}V = V|C|^{1/2}, \quad |C^\dagger|^{1/2}V|C|^{1/2} = C.$$

The polar operators lie in the von Neumann algebra generated by $\mathcal{A}$, and hence commute with B . Expanding yields

$$P_{C,B}^\dagger P_{C,B} = |C^\dagger| + |C| - CB - B^\dagger C^\dagger.$$

No inverse and no unitary extension of V has been used. $\square$

The second ingredient is a scalar equality calculation; its elementary parity-split proof is included in Appendix A.

Lemma 3.3 (Roots-of-unity extremum). *For $z \in \mathbb{T}$,*

$$\sum_{y=0}^{d-1} |1 + \omega^y z| \leq M_d. \quad (4)$$

Equality holds exactly when

$$z^d = (-1)^{d-1}. \quad (5)$$

Proof of Theorem 3.1. Set

$$L_y = A_0 + \omega^y A_1, \quad U = A_0^\dagger A_1, \quad M_y = \mathbf{1} + \omega^y U.$$

Then $L_y = A_0 M_y$. The operator M_y is normal because

$$M_y^\dagger M_y = M_y M_y^\dagger = 2\mathbf{1} + \omega^y U + \omega^{-y} U^\dagger.$$

Consequently

$$|L_y| = |M_y|, \quad |L_y^\dagger| = A_0 |M_y| A_0^\dagger.$$

Put

$$F_d(U) = \sum_{y=0}^{d-1} |\mathbf{1} + \omega^y U|.$$

Continuous functional calculus for the unitary U , applied to Lemma 3.3, gives $0 \leq F_d(U) \leq M_d \mathbf{1}$. Summing Lemma 3.2 now gives

$$\mathcal{I}_d \leq \frac{1}{2} (F_d(U) + A_0 F_d(U) A_0^\dagger) \leq M_d \mathbf{1}.$$

This uses only cross-party commutation, not a tensor-product factorization. The finite order- d strategy rechecked in Appendix B attains M_d . The inclusions among q, qa, qc then prove all three equalities. $\square$

For later equality arguments it is useful to retain the complete gap. If $L_y = V_y |L_y|$, define $P_y = P_{L_y, B_y}$ and $G = (M_d \mathbf{1} - F_d(U))^{1/2}$. Then

$$\boxed{M_d \mathbf{1} - \mathcal{I}_d = \frac{1}{2} \sum_y P_y^\dagger P_y + \frac{1}{2} G^\dagger G + \frac{1}{2} (G A_0^\dagger)^\dagger (G A_0^\dagger).} \quad (6)$$

This is an operator identity, not a finite-level NPA certificate. At the operator level, scalar functional calculus also gives

$$F_d(U) = M_d \mathbf{1} \iff U^d = (-1)^{d-1} \mathbf{1}.$$

For an individual maximizing vector $|\psi\rangle$, however, the exact conclusion from (6) is only

$$P_y|\psi\rangle = 0 \quad (0 \leq y < d), \quad G|\psi\rangle = 0, \quad GA_0^\dagger|\psi\rangle = 0.$$

We do not promote these support-level conditions to a global relation for U , or to a classification of all maximizing realizations.

Corollary 3.4 (First augmented family). *For*

$$\overline{\mathcal{I}}_d = \mathcal{I}_d + \text{Re}(A_0 B_d)$$

with a further order- d Bob observable,

$$\beta_q(\overline{\mathcal{I}}_d) = \beta_{qa}(\overline{\mathcal{I}}_d) = \beta_{qc}(\overline{\mathcal{I}}_d) = M_d + 1.$$

Proof. The upper bound follows from Theorem 3.1 and $\text{Re}(A_0 B_d) \leq \mathbf{1}$. The finite strategy in Appendix B, with the aligned added observable, attains equality. $\square$

4 Equality phases and permutation blindness

The scalar equality set in (5) is a set, not an ordering. A Bell score built only from first powers can therefore fail to record how those phases are arranged along a weighted cycle. The following general statement isolates the exact sufficient conditions. It does not claim to classify every maximizer of every polar-linear functional.

Let R be a finite coefficient index set, let $g_r(z) = \alpha_r + \beta_r z$, and put

$$f(z) = \sum_{r \in R} |g_r(z)|, \quad M = \max_{z \in \mathbb{T}} f(z).$$

For order- d observables define

$$\mathcal{J} = \sum_{r \in R} \text{Re}[(\alpha_r A_0 + \beta_r A_1) B_r] + \text{Re}(A_0 B_*). \quad (7)$$

Theorem 4.1 (Conditional phase-permutation theorem). *Suppose there are labeled phases $z_j, s_{rj} \in \mathbb{T}$, $0 \leq j < d$, such that*

$$f(z_j) = M, \quad \prod_j z_j = 1,$$

and, for every r ,

$$s_{rj} |g_r(z_j)| = g_r(z_j), \quad \prod_j s_{rj} = 1.$$

At a zero of g_r , s_{rj} may be any unitary extension phase. For every permutation κ of the labels, set on $\mathbb{C}^d$

$$\begin{aligned} A_0 &= X, & A_1 &= X \text{diag}(z_{\kappa_j}), \\ B_r &= \overline{X \text{diag}(s_{r, \kappa_j})}, & B_* &= X, \end{aligned} \quad (8)$$

where $X|j\rangle = |j+1 \bmod d\rangle$, and use $|\Phi_d\rangle = d^{-1/2} \sum_j |j, j\rangle$. Then:

- (i) every displayed observable is unitary, has d -th power $\mathbf{1}$, and has the full simple d -th-root spectrum;*

- (ii) the strategy attains the operator upper bound $M + 1$ for (7); and
- (iii) all local first moments and every complex first-harmonic correlator $\langle A_x B_r \rangle$, $\langle A_x B_* \rangle$ are independent of κ .

Proof. With $U = A_0^\dagger A_1$ and $L_r = A_0 g_r(U) = V_r |L_r|$, Lemma 3.2 and functional calculus give

$$(M + 1)\mathbf{1} - \mathcal{J} = \frac{1}{2} \sum_r P_r^\dagger P_r + \frac{1}{2} R_*^\dagger R_* + \frac{1}{2} (H + A_0 H A_0^\dagger),$$

where

$$P_r = |L_r^\dagger|^{1/2} - V_r |L_r|^{1/2} B_r, \quad R_* = \mathbf{1} - A_0 B_*, \quad H = M\mathbf{1} - f(U) \geq 0.$$

Thus $M + 1$ is an operator upper bound.

For arbitrary nonzero weights t_j ,

$$(X \operatorname{diag}(t_j))^d = \left(\prod_j t_j \right) \mathbf{1}, \tag{9}$$

and its characteristic polynomial is $t^d - \prod_j t_j$. The product hypotheses therefore prove (i). Moreover,

$$\alpha_r A_0 + \beta_r A_1 = X \operatorname{diag}(g_r(z_{\kappa_j}))$$

has unitary polar extension $X \operatorname{diag}(s_{r,\kappa_j})$. The maximally-entangled trace identity

$$\langle \Phi_d | R \otimes S | \Phi_d \rangle = \frac{1}{d} \operatorname{Tr}(RS^T)$$

annihilates every factor in the displayed gap, proving (ii). Finally,

$$\begin{aligned} \langle A_0 B_r \rangle &= \frac{1}{d} \sum_j \overline{s_{rj}}, & \langle A_1 B_r \rangle &= \frac{1}{d} \sum_j z_j \overline{s_{rj}}, \\ \langle A_0 B_* \rangle &= 1, & \langle A_1 B_* \rangle &= \frac{1}{d} \sum_j z_j. \end{aligned}$$

These are symmetric sums of the labeled data. Every local observable is a weighted shift and has zero trace. This proves (iii). $\square$

The theorem explains both its power and its limit. It gives a concrete permutation orbit within a maximizing face, but it says nothing about maximizers whose relative spectrum is different, has different multiplicities, acts reducibly, or uses a kernel outside the stated phase data. Claims about the complete maximizing face require separate rigidity analysis.

5 Exact nonuniform maximizers of the first family

Put

$$\delta_d = \begin{cases} 0, & d \text{ odd}, \\ 1, & d \text{ even}, \end{cases} \quad z_k = \exp\left(\frac{\pi i(2k + \delta_d)}{d}\right).$$

These are precisely the roots in (5). For $0 \leq y, k < d$, define

$$s_{y,k} = \frac{1 + \omega^y z_k}{|1 + \omega^y z_k|}.$$

The denominator never vanishes. The polynomial $t^d - (-1)^{d-1}$ gives

$$\prod_k z_k = 1, \quad \prod_k (1 + \omega^y z_k) = 2, \quad \prod_k s_{y,k} = 1. \quad (10)$$

Thus Theorem 4.1 applies with $g_y(z) = 1 + \omega^y z$.

For every permutation κ , the explicit first-family strategy is

$$\begin{aligned} A_0 &= X, & A_1 &= XZ_\kappa, & Z_\kappa &= \text{diag}(z_{\kappa_j}), \\ V_y &= X \text{diag}(s_{y,\kappa_j}), & B_y &= \overline{V_y}, & B_d &= X. \end{aligned} \quad (11)$$

Each observable defines a rank-one projective d -outcome measurement and

$$\langle \overline{\mathcal{I}}_d \rangle = M_d + 1.$$

The Bell-visible correlators are independent of the ordering. Explicitly, with $\alpha_d = \csc(\pi/(2d))$,

$$\langle A_0 B_y \rangle = \frac{\alpha_d}{d}, \quad \langle A_1 B_y \rangle = \frac{\alpha_d \omega^{-y}}{d}, \quad \langle A_0 B_d \rangle = 1, \quad \langle A_1 B_d \rangle = 0. \quad (12)$$

The cyclic ordering $\kappa_j = j$ reproduces the canonical first-harmonic behavior. Other orderings can change higher powers and full probabilities.

To calculate the designated pair (A_1, B_d) , set

$$q_0 = 1, \quad q_{j+1} = z_{\kappa_j} q_j, \quad \widehat{q}_m = \sum_{j=0}^{d-1} q_j \omega^{mj}. \quad (13)$$

The product condition makes the sequence cyclic. Direct diagonalization of the two weighted shifts gives

$$p_\kappa(a, b \mid 1, d) = \frac{|\widehat{q}_{-(a+b)}|^2}{d^3}. \quad (14)$$

Parseval gives $\sum_m |\widehat{q}_m|^2 = d^2$. Consequently both local marginals are exactly $1/d$, while the joint table is uniform if and only if every squared Fourier magnitude equals d .

Theorem 5.1 (Biased exact maximizers). *For every $d \geq 4$, choose the final-two swap*

$$\kappa = (0, 1, \dots, d-3, d-1, d-2). \quad (15)$$

The strategy (11) attains $M_d + 1$, has uniform local marginals, and has a nonuniform target table. With one-dimensional Eve,

$$G(AB \mid 1, d, E) \geq \frac{1}{d^2} + \frac{2 \sin(\pi/d) \sin(3\pi/d)}{d^2(d-1)} > \frac{1}{d^2}. \quad (16)$$

Proof. The second cyclic autocorrelation of (13) is

$$R_2 = \sum_j q_{j+2} \overline{q_j} = \sum_j z_{\kappa_j} z_{\kappa_{j+1}}.$$

It vanishes for the cyclic ordering. The swap (15) gives

$$R_2 = (z_{d-1} - z_{d-2})(z_{d-3} - z_0), \quad |R_2| = 4 \sin(\pi/d) \sin(3\pi/d) > 0. \quad (17)$$

Hence the Fourier magnitudes are not all d .

For the quantitative estimate put $x_m = |\hat{q}_m|^2 - d$ and $\Delta = \max_m x_m$. Fourier inversion at lag two gives

$$R_2 = \frac{1}{d} \sum_m x_m \omega^{-2m},$$

where the constant term vanishes for $d \geq 4$. Since the nonzero mean-zero sequence (x_m) has at most $d-1$ positive entries,

$$d|R_2| \leq \sum_m |x_m| = 2 \sum_{x_m > 0} x_m \leq 2(d-1)\Delta.$$

With trivial Eve, always guessing a most likely pair gives $G = (d+\Delta)/d^3$. Substituting (17) proves (16). $\square$

For $d = 2, 3$, every ordering in this particular permutation family is Fourier-flat. The theorem therefore neither proves nor disproves Bell-value rigidity in those dimensions.

6 Extension to the second augmented cyclic family

The second reduced family of Ref. [1] has a different Alice measurement set but compresses the same first-family Bob observables. Put $\eta = e^{\pi i/d}$, so $\omega = \eta^2$, and define

$$\lambda_\ell = \frac{(-1)^{\ell-1} \eta^{\ell(\ell-1)}}{d \sin(\pi(\ell - \frac{1}{2})/d)}, \quad 0 \leq \ell < d, \quad (18)$$

where $(-1)^{-1} = -1$ at $\ell = 0$. For Bob's observables in (11), let

$$\hat{B}_\ell = \sum_{y=0}^{d-1} \omega^{\ell y} B_y$$

and define the Hermitian functional

$$\mathcal{F}_d = \sum_{\ell=0}^{d-1} \operatorname{Re}(\overline{\lambda_\ell} A_\ell \hat{B}_\ell), \quad \overline{\mathcal{F}}_d = \mathcal{F}_d + \operatorname{Re}(A_0 B_d). \quad (19)$$

The source SOS is short and load-bearing. Fourier orthogonality and unitarity of the B_y 's give

$$\sum_\ell \hat{B}_\ell^\dagger \hat{B}_\ell = d^2 \mathbf{1}, \quad \sum_\ell |\lambda_\ell|^2 = 1.$$

Thus, for $P_\ell = d\lambda_\ell \mathbf{1} - A_\ell \widehat{B}_\ell$,

$$d\mathbf{1} - \mathcal{F}_d = \frac{1}{2d} \sum_{\ell=0}^{d-1} P_\ell^\dagger P_\ell. \quad (20)$$

The identity requires only cross-party commutation, so the source value also has the immediate commuting-operator reading

$$\beta_q(\mathcal{F}_d) = \beta_{qa}(\mathcal{F}_d) = \beta_{qc}(\mathcal{F}_d) = d, \quad \beta_q(\overline{\mathcal{F}}_d) = \beta_{qa}(\overline{\mathcal{F}}_d) = \beta_{qc}(\overline{\mathcal{F}}_d) = d + 1.$$

Theorem 6.1 (Permutation maximizers of the second family). *For every permutation κ , define*

$$r_\ell = \eta^{-\ell(\ell-1+\delta_d)}, \quad D_\ell = r_\ell X \operatorname{diag}(\omega^{-\ell\kappa_j}), \quad A_\ell = \overline{D}_\ell. \quad (21)$$

Together with the B_y 's in (11), these are order- d observables and

$$\widehat{B}_\ell = d\lambda_\ell D_\ell. \quad (22)$$

They annihilate every factor in (20). With $B_d = X$, the augmented value is $d + 1$. For the swap (15), the target pair (A_1, B_d) has exactly the table (14) and the gap (16) for every $d \geq 4$. The complete matrix of complex first-harmonic correlators is independent of κ . Under the alternative source-appendix convention that adjoints all Bob observables, the target table is the same up to the harmless outcome inversion $b \mapsto -b$.

Proof. For $z_t = \eta^{2t+\delta_d}$, write

$$s_{0,t} = \varepsilon_t \eta^{t+\delta_d/2}, \quad \varepsilon_t = \begin{cases} +1, & 2t + \delta_d < d, \\ -1, & 2t + \delta_d > d. \end{cases}$$

There is no equality case, and $s_{y,t} = s_{0,t+y \bmod d}$. The diagonal weight of $\widehat{B}_\ell$ at a label t is therefore $\omega^{-\ell t} S_\ell$, with

$$S_\ell = \sum_t \omega^{\ell t} \overline{s_{0,t}}.$$

Let $n = (d + 1 - \delta_d)/2$ and $u_\ell = \eta^{2\ell-1}$. Since $u_\ell^d = -1$, two geometric sums give

$$S_\ell = \eta^{-\delta_d/2} \left(\sum_{t=0}^{n-1} u_\ell^t - \sum_{t=n}^{d-1} u_\ell^t \right) = d\lambda_\ell r_\ell.$$

This is (22), including its phase.

By (9) and $\sum_j \kappa_j = d(d-1)/2$,

$$D_\ell^d = (-1)^{\ell(\ell-1+\delta_d)+\ell(d-1)} \mathbf{1} = \mathbf{1},$$

because $\ell(\ell + d - 2 + \delta_d)$ is even. Equation (22) gives

$$P_\ell |\Phi_d\rangle = d\lambda_\ell (\mathbf{1} - \overline{D}_\ell \otimes D_\ell) |\Phi_d\rangle = 0.$$

Also $A_0 = X$, so $X \otimes X$ fixes $|\Phi_d\rangle$. Finally

$$A_1 = X \operatorname{diag}(\eta^{2\kappa_j+\delta_d}) = X \operatorname{diag}(z_{\kappa_j}),$$

which identifies the target table with (14). Each first-harmonic correlator is a sum over the complete unordered root set, and is therefore permutation-independent. $\square$

The theorem uses the source SOS to prove global optimality; merely annihilating a candidate collection of operators would not have sufficed. What is common to both families is the root ordering, the Bob cycle, and the target table. The second family additionally requires the exact Fourier phases in (21). No statement is made here about a complete classification of its maximizing face.

7 Randomness interpretation

Define the value-conditioned guessing quantity for either augmented family by

$$G_{\text{val}}(d) = \sup_{\mathcal{S}: \langle \mathcal{B}_d \rangle = \beta(\mathcal{B}_d)} G_{\mathcal{S}}(AB \mid 1, d, E),$$

where $\mathcal{B}_d$ is $\overline{\mathcal{I}}_d$ or $\overline{\mathcal{F}}_d$. Theorems 5.1 and 6.1 imply

$$G_{\text{val}}(d) \geq \frac{1}{d^2} + \frac{2 \sin(\pi/d) \sin(3\pi/d)}{d^2(d-1)} > \frac{1}{d^2} \quad (d \geq 4). \quad (23)$$

This is a lower bound on the worst case over the maximizing face. We do not claim that the final-two swap is worst possible.

The scope can be stated in one sentence:

The maximum scalar Bell value alone does not identify the canonical behavior or certify maximal global randomness across the entire maximizing face.

It is equally important to state what does not follow. The result does not dispute a calculation that fixes the complete canonical behavior, and it does not show that the canonical strategy lacks maximal global randomness. In fact, the numerical program described in Appendix B.1 of Ref. [1] reconstructs the full distribution of the canonical strategy before optimizing Eve's guessing probability. The root-swapped strategy has different higher Fourier correlators and therefore does not satisfy those full-behavior constraints.

A full-behavior SDP may thus return $G = 1/d^2$ for the canonical table while (23) holds for the scalar maximum. Passing from the first statement to the second would require uniqueness of the maximizing behavior, self-testing strong enough to control the target PVMs, or other statistics that exclude the phase-permuted realization. The source's $d = 3$ self-testing theorem for the second family is outside our $d \geq 4$ construction and is not challenged.

Corollary 7.1 (No value-only endpoint robustness). *For either augmented family and every $d \geq 4$, there is no valid bound for all strategies of the form*

$$G(AB \mid E) \leq \frac{1}{d^2} + f_d(\varepsilon), \quad f_d(\varepsilon) \longrightarrow 0,$$

where ε is only the Bell deficit from the maximum. More precisely, the asserted bound is required to hold for every strategy whose deficit is at most ε .

Proof. The biased exact maximizer has deficit zero, hence is feasible for every $\varepsilon \geq 0$, and has the fixed positive gap in (23). Bounds that use additional observed correlators or the complete behavior are not excluded. $\square$

8 Setting-complexity consequences

The permutation mechanism is a design warning, not a universal lower bound on settings. One universal baseline is nevertheless immediate.

Proposition 8.1 (One-input baseline). *If either party has only one input, every finite-output nonsignalling behavior is local and admits a compatible finite-dimensional pure projective realization in which Eve guesses both outputs perfectly. Hence at least two inputs per party are necessary for any private global randomness, even when the complete behavior is fixed.*

Proof. Suppose Alice has one input. Nonsignalling makes $p(a) = \sum_b p(a, b | y)$ independent of y . For $p(a) > 0$, let $r_y(b | a) = p(a, b | y)/p(a)$, and choose an arbitrary conditional distribution when $p(a) = 0$. The hidden variable

$$\lambda = (a, b_1, \dots, b_{m_B}), \quad \mu(\lambda) = p(a) \prod_y r_y(b_y | a)$$

reproduces the whole behavior by outputting the stored entries. The pure state

$$\sum_{\lambda} \sqrt{\mu(\lambda)} |\lambda\rangle_A |\lambda\rangle_B |\lambda\rangle_E$$

with diagonal grouping PVMs realizes it. Eve reads λ and guesses perfectly. The other case is symmetric. $\square$

This is a standard locality fact rather than a novelty claim. For $d = 2$, the $(2, 2)$ baseline is attainable by a known tilted Bell score [10]. For $d \geq 3$, neither this paper nor the phase-permutation theorem determines the minimum number of settings. A current all-dimensional construction with projective generation measurements uses two Alice settings and $d^2 + 1$ Bob settings [14].

Appendix D records two narrower observations retained from the setting-complexity study: the standard two-input qudit self-test has no uniform joint setting pair, and one natural separately bounded third-setting repair cannot expose its common computational mutually unbiased basis. The obstruction does not rule out a genuinely joint SOS, another MUB, another self-test, or an arbitrary $(2, 3, d, d)$ Bell functional.

9 Discussion and open problems

The exact value theorem and the biased maximizers expose a structural distinction between optimality and rigidity. The first operator controls a sum of moduli of first-order linear functions of the relative unitary. Its score is sharp enough to determine the allowable scalar phases but not their cyclic order. That missing order is invisible to the Bell score and visible to the target output distribution.

Several questions are now sharply posed.

1. Characterize the complete maximizing faces of both augmented families, including reducible and commuting-operator realizations.
2. Determine the exact worst-case guessing probability at the Bell maximum; the bound (23) need not be tight.
3. Resolve the $d = 2, 3$ maximizing faces for the first augmented family. The permutation orbit studied here is uniform in those dimensions.

4. Classify which phase permutations are related by local isometries and output relabelings, and which produce genuinely different full behaviors.
5. Find a low-setting modification whose Bell score controls the missing higher Fourier data, and determine whether it is rigid or robustly self-testing.
6. Develop robust bounds conditioned on a minimal set of extra correlators, rather than either one scalar score or the entire behavior.

These problems are natural opportunities for collaboration with the originating authors and the broader Bell-inequality community. No external review or collaboration is claimed here.

Data, code, and historical versions

The accompanying repository contains the TeX source, exact and numerical verifiers, machine-readable certificates, claims ledger, proof-dependency map, fresh adversarial and priority audits, and integrity hashes. One command replays the mathematical checks, the two independent $d = 4$ exact implementations, the manuscript build, and the website validation. The three earlier standalone papers and their original PDFs, sources, hashes, and Git history remain available as historical artifacts; they are not silently replaced by this synthesis.

AI-assistance disclosure

Generative-AI systems assisted with proof exploration, adversarial testing, code, literature-search organization, and drafting under Alec Kriebel's direction. Alec Kriebel is the sole named author and takes responsibility for the manuscript and release. AI output was not treated as evidence: claims were retained only when accompanied by a written proof and replayable checks. No originating author or outside researcher was contacted.

A Proof of the scalar extremum

Write $z = e^{2is}$, $h = \pi/d$, and $a = h/2$. Half of the sum in (4) is

$$G_d(s) = \sum_{y=0}^{d-1} |\cos(s + yh)|,$$

which has period h .

If $d = 2m + 1$, take $-a \leq s \leq a$ and reindex by $k = -m, \dots, m$. Every angle lies in $[-\pi/2, \pi/2]$, so

$$G_d(s) = \sum_{k=-m}^m \cos(s + kh) = \csc(a) \cos s.$$

The maximum occurs exactly at $s \equiv 0 \pmod{h}$, equivalently $z^d = 1$.

If $d = 2m$, take $0 \leq s \leq h$. The first m cosines are nonnegative and the last m are nonpositive. Two finite sums give

$$G_d(s) = \sum_{y=0}^{m-1} \cos(s + yh) - \sum_{y=m}^{2m-1} \cos(s + yh) = \csc(a) \cos(s - a).$$

The maximum occurs exactly at $s \equiv a \pmod{h}$, equivalently $z^d = -1$. Multiplying by two proves Lemma 3.3, including the equality set.

B Rechecking the canonical attaining strategy

This appendix verifies the lower strategy supplied in Ref. [1] in a polar form. On $\mathbb{C}^d$, let

$$Z|j\rangle = \omega^j|j\rangle, \quad X|j\rangle = |j+1 \bmod d\rangle, \quad A_0 = Z, \quad A_1 = X.$$

Set

$$W_y = \omega^y Z^\dagger X, \quad H_y = |\mathbf{1} + W_y|, \quad Q_y = H_y^{-1}(\mathbf{1} + W_y^\dagger)Z^\dagger, \quad B_y = Q_y^T.$$

The weighted-shift action gives

$$W_y^d = (-1)^{d-1} \mathbf{1}, \quad \text{spec}(W_y) = \{\lambda : \lambda^d = (-1)^{d-1}\}.$$

The spectrum is simple, and -1 is absent; hence H_y is invertible. For $L_y = Z + \omega^y X = Z(\mathbf{1} + W_y)$,

$$L_y = \underbrace{Z(\mathbf{1} + W_y)H_y^{-1}}_{V_y} H_y$$

is the polar decomposition, and $Q_y = V_y^\dagger$ is unitary.

It remains to check the measurement order. On the displayed spectrum put $p(z) = (1+z)/|1+z|$. Since $Z^\dagger W_y Z = \omega^{-1} W_y$,

$$Q_y^d = \prod_{r=0}^{d-1} p(\omega^{-r} W_y)^\dagger.$$

For each eigenvalue λ , the corresponding scalar equals

$$\frac{\overline{\prod_r (1 + \omega^{-r} \lambda)}}{\prod_r |1 + \omega^{-r} \lambda|} = \frac{\overline{1 - (-\lambda)^d}}{|1 - (-\lambda)^d|} = 1.$$

Thus $B_y^d = \mathbf{1}$. The weighted-cycle characteristic polynomial shows that each d -th root occurs.

Finally,

$$\langle \Phi_d | L_y \otimes B_y | \Phi_d \rangle = \frac{1}{d} \text{Tr}(L_y V_y^\dagger) = \frac{1}{d} \text{Tr}(H_y).$$

The spectrum of W_y is the scalar equality set, so summing over y gives $\langle \mathcal{I}_d \rangle = M_d$. Taking the extra Bob observable $B_d = Z^\dagger$ gives $\langle Z \otimes Z^\dagger \rangle = 1$, proving Corollary 3.4. This appendix claims no new attaining strategy.

C The exact four-outcome certificate

Let $d = 4$, $\zeta = e^{\pi i/8}$, and $\kappa = (0, 1, 3, 2)$. Then

$$A_0 = X, \quad A_1 = X \text{diag}(\zeta^2, \zeta^6, \zeta^{14}, \zeta^{10}), \quad B_4 = X.$$

The first-family polar unitaries $V_y = X \operatorname{diag}(\zeta^{e_{y,0}}, \dots, \zeta^{e_{y,3}})$ are encoded by

y	$e_{y,0}$	$e_{y,1}$	$e_{y,2}$	$e_{y,3}$	
0	1	3	15	13	$B_y = \overline{V_y}$.
1	3	13	1	15	
2	13	15	3	1	
3	15	1	13	3	

These are exact monomial matrices over $\mathbb{Q}(\zeta_{16})$, not numerical polar decompositions.

The recurrence (13) gives

$$q = (1, \zeta^2, -1, \zeta^6), \quad (|\widehat{q}_m|^2)_{m=0}^3 = (2, 6, 2, 6).$$

For both augmented families,

$$p(a, b \mid 1, 4) = \begin{cases} 1/32, & a + b \text{ even,} \\ 3/32, & a + b \text{ odd,} \end{cases} \quad G = 3/32 > 1/16. \quad (24)$$

One exact verifier reconstructs (24) from projectors and a separate Fourier calculation. An independent exact implementation reconstructs the source coefficients $\lambda_{y,k}$, proves $\widehat{B}_\ell = 4\lambda_\ell D_\ell$, checks all fourth-power relations, expands the full second-family SOS, and verifies values $M_4 + 1$ and 5.

D Scoped low-setting observations

The standard two-input maximally-entangled-qudit self-test [9] uses Fourier-phase bases. Up to input swaps, conjugation, transposition, and outcome relabeling, its four ideal tables have the form

$$p(a, b \mid x, y) = \frac{\sin^2(\pi \delta_{xy})}{d^3 \sin^2(\pi(a - b + \delta_{xy})/d)}, \quad (\delta_{xy}) = \begin{pmatrix} 1/4 & 3/4 \\ -1/4 & 1/4 \end{pmatrix}. \quad (25)$$

Every table has

$$p_{\max} = \frac{1}{2d^3 \sin^2(\pi/(4d))} > \frac{1}{d^2},$$

so no ideal input pair is uniform. Perfectly anchoring a third Bob setting to either Alice basis leaves a cross table with

$$p_{\max}^{\text{cross}} = \frac{1}{d^3 \sin^2(\pi/(2d))} > \frac{1}{d^2} \quad (d \geq 3).$$

The exception $d = 2$ is precisely the mutually-unbiased qubit case.

There is a broader but still scoped obstruction. Let F be the Fourier matrix, $\rho = e^{\pi i/d}$, and $R = \operatorname{diag}(1, \rho, \dots, \rho^{d-1})$. The real Hermitian operator system generated by the two ideal Alice PVMs is

$$\mathcal{S} = \{F \operatorname{diag}(x) F^\dagger + R F \operatorname{diag}(y) F^\dagger R^\dagger : x, y \in \mathbb{R}^d\}.$$

Proposition D.1 (Computational-MUB exposure obstruction). *If $K \in \mathcal{S}$ has a computational vector e_r as an eigenvector with eigenvalue κ , then either $K = \kappa \mathbf{1}$, or κ is neither the largest nor the smallest eigenvalue of K . Consequently, no nontrivial separately bounded term*

$$\mathcal{L} = \sum_b K_b \otimes Q_b, \quad K_b \in \mathcal{S},$$

can expose the computational PVM on $|\Phi_d\rangle$ through a coefficientwise spectral upper bound.

Proof. Write $K = C + RC'R^\dagger$ with C, C' Hermitian circulants. Comparing the column- r eigenvector equations with their Hermitian conjugates forces all but the first and last $s = \min(r, d - 1 - r)$ cyclic diagonals to vanish. After grouping the first s , middle $d - 2s$, and last s coordinates,

$$K - \kappa \mathbf{1} = \begin{pmatrix} 0 & 0 & T \\ 0 & 0 & 0 \\ T^\dagger & 0 & 0 \end{pmatrix},$$

where T is upper-triangular Toeplitz. Its nonzero eigenvalues occur in pairs $\pm \sigma_j(T)$. Hence a nonscalar K has spectrum on both sides of κ .

To saturate a coefficientwise upper bound on the maximally entangled state with computational Q_b , every computational outcome has nonzero weight and the corresponding e_b must be a maximal eigenvector of K_b . The first part makes that impossible unless all relevant K_b are scalar, in which case the term constrains no measurement. $\square$

Proposition D.1 rules out only this separately bounded local- spectral route. It is not a no-go theorem for $(2, 3, d, d)$, and it is not used in any cyclic-family conclusion.

E Source conventions and claim boundary

Our first-family definition (1) follows the displayed main-text operator definition in Ref. [1]. A nearby appendix writes one Bob adjoint differently while its subsequent calculation follows the main convention. Replacing every Bob observable by its adjoint transports between the two notations, but mixing them within one formula does not. We therefore state every adjoint explicitly and test the Fourier orientation.

For the second family, (19) matches the Hermitian content of the source definition and its corrected $1/(2d)$ SOS factor in the latest version. The source text adjacent to the canonical construction briefly omits Alice's entrywise conjugation, while its theorem includes it. Our choice $A_\ell = \overline{D_\ell}$ is the one that annihilates the displayed SOS on $|\Phi_d\rangle$.

If instead every Bob observable in the source appendix is written with an adjoint, set $B'_y = B_y^\dagger$ and $B'_d = B_d^\dagger$. This is Bob's outcome inversion: it sends $p(a, b)$ to $p(a, -b)$ while preserving the maximum, nonuniformity, and guessing probability. It cannot be mixed termwise with the no-adjoint convention used in (19).

The source's numerical guessing-probability table explicitly fixes the full canonical distribution. Our scalar-value conclusion should therefore not be read as a criticism of that calculation. The literal maximal-violation conjecture and the fixed-behavior numerical experiment have different logical hypotheses; the present construction separates them.

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